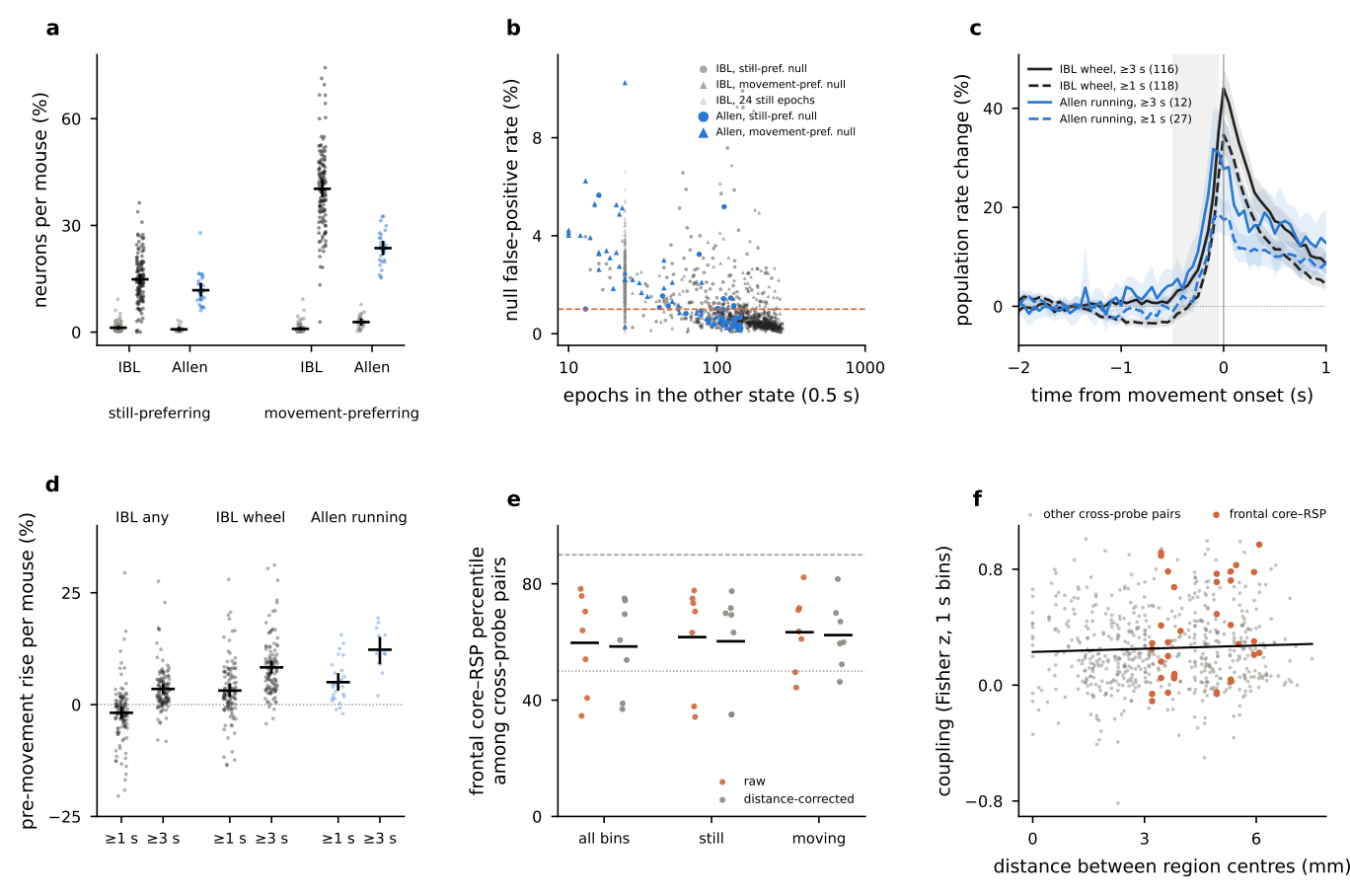

a
neurons per mouse (%)
60
30
0
IBL
Allen
IBL
Allen
still-preferring
movement-preferring
b
null false-positive rate (%)
8
4
0
IBL, still-pref. null
IBL, movement-pref. null
IBL, 24 still epochs
Allen, still-pref. null
Allen, movement-pref. null
10
100
1000
epochs in the other state (0.5 s)
c
population rate change (%)
40
20
0
IBL wheel, ≥3 s (116)
IBL wheel, ≥1 s (118)
Allen running, ≥3 s (12)
Allen running, ≥1 s (27)
−2
−1
0
1
time from movement onset (s)
d
pre-movement rise per mouse (%)
25
0
−25
IBL any
IBL wheel
Allen running
≥1 s
≥3 s
≥1 s
≥3 s
≥1 s
≥3 s
e
frontal core–RSP percentile among cross-probe pairs
80
40
0
raw
distance-corrected
all bins
still
moving
f
other cross-probe pairs
frontal core–RSP
coupling (Fisher z, 1 s bins)
0.8
0.0
−0.8
0
3
6
distance between region centres (mm)